



اَبُو سَيِّدٍ تَيْكُوْلُو كِي مَبَارَا
UNIVERSITI
TEKNOLOGI
MARA

Kolej
Pengajian Pengkomputeran,
Informatik dan Matematik

**UNIVERSITI TEKNOLOGI MARA (UiTM) CAWANGAN MELAKA
KAMPUS JASIN**

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**ICT502
DATABASE ENGINEERING**

**PROJECT TITLE:
GOVERNMENT CLINIC CHENG APPOINTMENT
BOOKING SYSTEM**

PREPARED BY:

STUDENT NAME	STUDENT ID
1. MAISARAH YASMIN BINTI ISHAK	2024807316
2. MUHAMMAD DANIEL BIN MOHD SAHID	2024213154
3. NUR FARA'AIN ALIYAH BINTI IMRAN	2024211146
4. PUTRI ZULAIKHA BINTI RASHIDI	2024441648
5. NUR SYAKILLA ASYIQIN BINTI HASAN	2024425838

PREPARED FOR:

SIR MUHAMMAD HAMIZ BIN MOHD RADZI

SUBMISSION DATE:

20th JANUARY 2025

TABLE OF CONTENT

1.0 COMPANY BACKGROUND.....	1
2.0 PROBLEM STATEMENT.....	2
3.0 OBJECTIVES.....	3
4.0 FLOW CHART.....	4
5.0 SQL QUERIES.....	5
6.0 SYSTEM DEVELOPMENT SAMPLE SCREEN.....	10
USER LOGIN (ADMIN, DOCTOR AND NURSE).....	10
DASHBOARD (ADMIN, DOCTOR AND NURSE).....	11
ADMIN DASHBOARD.....	11
7.0 CONCLUSION.....	28
8.0 APPENDIX.....	29
8.1 ENTITY RELATIONSHIP DIAGRAM.....	29
8.2 3NF RELATIONAL SCHEMA.....	30
8.3 DATA DICTIONARY.....	31
8.4 DATA DEFINITION LANGUAGE (DDL).....	33
8.5 DATA MANIPULATION LANGUAGE (DML).....	39
8.6 CD/FLASH DRIVE.....	46
9.0 REFERENCES.....	47
10.0 RUBRIC.....	48

1.0 COMPANY BACKGROUND

Government Clinic Cheng (GCC) refers to government-run healthcare facilities that are governed by the Jabatan Kesihatan Negeri Melaka (JKNM). These clinics are a part of the public healthcare system in Malaysia, which provides primary health care that is easily accessible and economically priced for all citizens of the nation. In Melaka, the government runs more than 33 clinics. These clinics, which are spread out over the districts of Jasin, Alor Gajah, and Melaka Tengah, provide outpatient, maternity, child health services, vaccinations, and basic diagnostics. They serve as the initial point of contact for many residents seeking medical attention, and they are crucial to disease control initiatives and public health campaigns.

GCC provides basic medical services with a focus on curative, rehabilitative, and preventive care in Taman Cheng Perdana, Melaka Tengah. It serves as an essential healthcare facility for those who live there. It is divided into three departments which are medical examination, dentistry, and public health. Additionally, the building has its own farmasi. Under the management of JKNM, GCC is dedicated to provide accessible healthcare services to the community, aligning with its vision of ensuring “Melaka residents healthy for life” and its mission to empower the people through promotion, prevention, treatment, and rehabilitation for quality living.

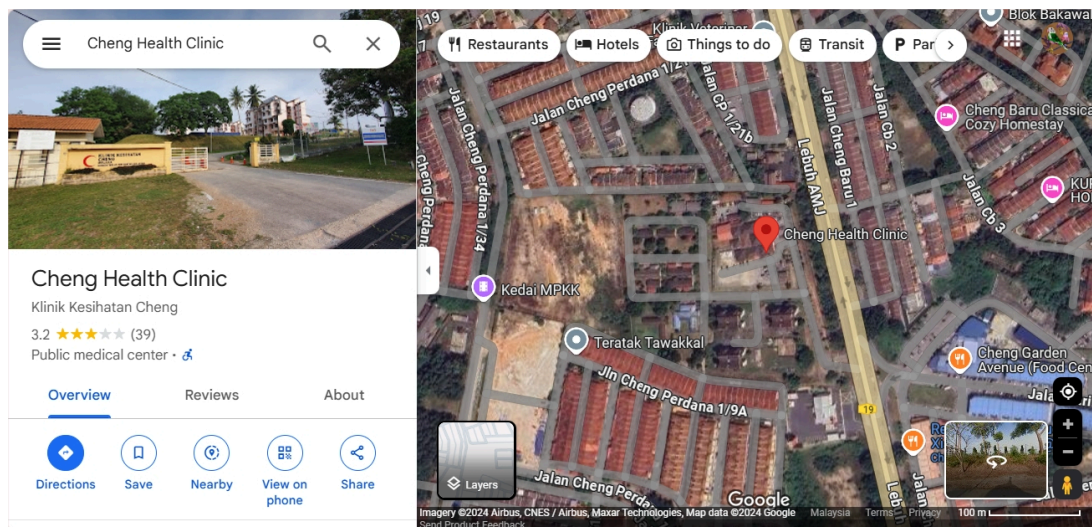


Figure 1.0: GCC in Google Maps

2.0 PROBLEM STATEMENT

GCC currently relies on a manual, file-based system to manage patient appointment records. This approach introduces several issues that affect the efficiency, accuracy, and security of healthcare services, including:

i) Manual management

The current manual registration process at the GCC leads to long waiting times and increases the workload of the staff. Everything is still done manually, from registration, booking appointments to taking medicine. Patients have to hand over their medical cards, wait in line, and everything is written on paper. This slows things down, makes more work for the staff, and leads to long waiting times, which can be frustrating for everyone involved.

ii) Difficulty in Managing Records

Currently, appointment records are stored manually, which increases the risk of errors, loss, and duplication. Updating records by hand can lead to inconsistencies, as some records may be updated while others are missed. This makes it difficult to maintain accurate and organized information.

iii) Lack of Real-Time Appointment Updates and Tracking

The current manual system gives a difficulty to the staff to quickly update or track the appointment. This leads to challenges such as double-booking, miscommunication, and confusion about the availability of appointment slots.

3.0 OBJECTIVES

In order to overcome the problems caused by a manual or file-based approach, the development process of this information system must achieve several objectives, which are:

i) To design a system that streamlines appointment booking

The system will feature a user-friendly interface, enabling staff to book appointments by viewing available dates and times. This will save more time and energy, eliminating the need for long wait times to schedule appointments in person, as well as reducing waiting times and lightening the scheduling workload for clinic staff, enhancing overall efficiency.

ii) To develop a system for comprehensive records

The system will play a vital role in maintaining detailed records of patient information, with a focus on appointment details. It will enhance processing and simplify workflows, allowing for quick and easy access to complete medical histories. This immediate access will empower healthcare providers to make informed clinical decisions, ultimately improving patient care.

iii) To test the functionality of the system

The system will be tested to ensure that it can effectively simplify the process of booking appointments with doctors, making it easier for the clinic staff. This will involve verifying that the system streamlines the booking process, reduces errors as much as possible, and improves efficiency compared to the manual or file-based approach. Successful testing will confirm that the system achieves its goal of simplifying appointment scheduling.

4.0 FLOW CHART

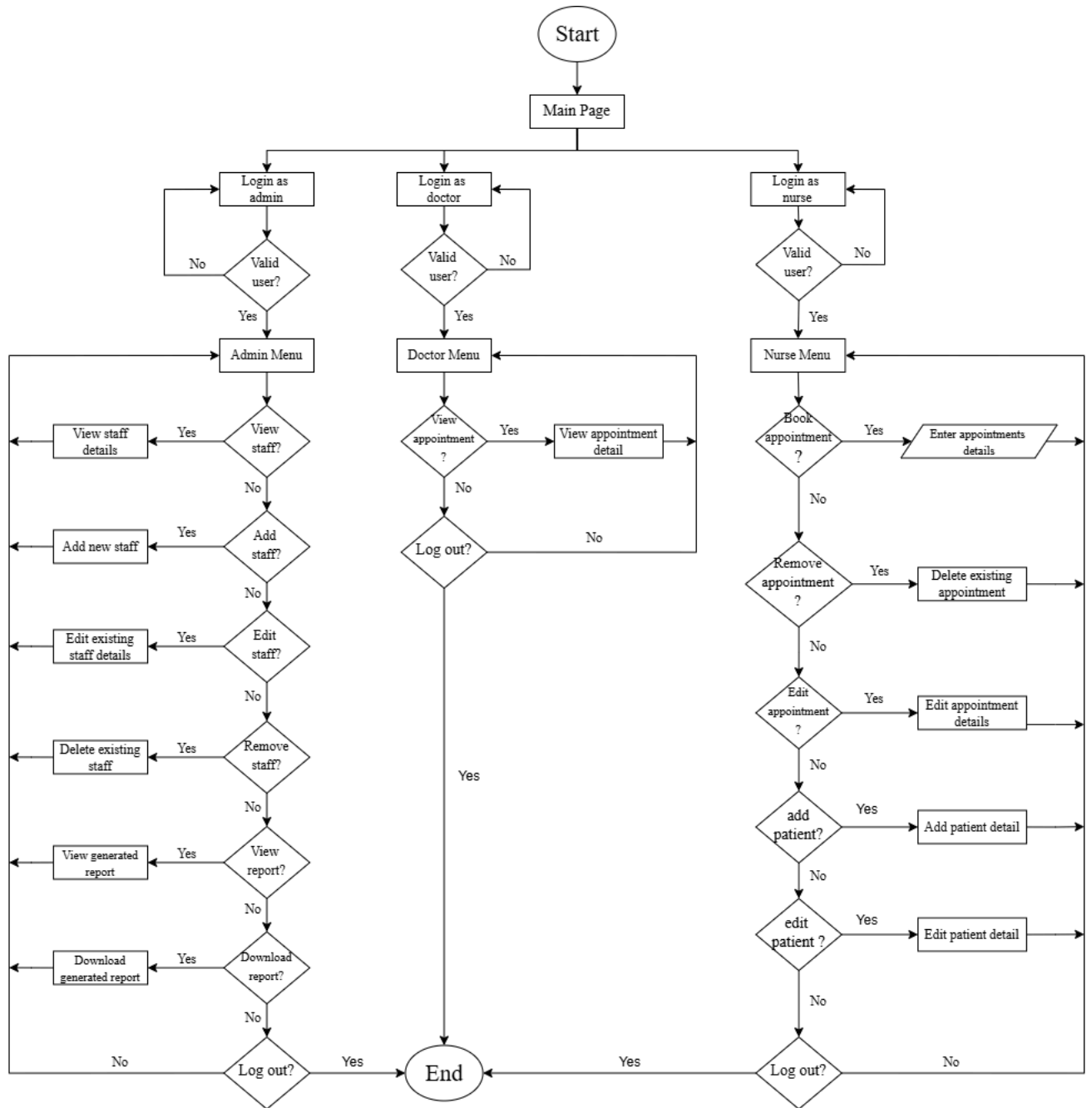


Figure 4.0: Flow Chart of The System

5.0 SQL QUERIES

1. FIND TOTAL OF PATIENTS HELP BY EACH DOCTOR (GROUP BY)

```
SELECT COUNT(PATIENT_ID) AS "TOTAL PATIENTS", S.STAFF_NAME
FROM APPOINTMENT APPT
JOIN STAFF S
ON S.STAFF_ID = APPT.STAFF_ID
GROUP BY S.STAFF_ID;
```

```
1 select count(patient_id) as "TOTAL PATIENTS", s.staff_name
2 from appointment appt
3 join staff s
4 on s.staff_id = appt.staff_id
5 group by s.staff_name;
6
```

Results	Explain	Describe	Saved SQL	History
TOTAL PATIENTS				STAFF_NAME
3				HAMIZAH
5				Daniel
9				MUHAMMAD
3				NOOR AMIRAH
2				Aqil

2. RETRIEVE ALL THE STAFF WHO IS DOCTOR

```
SELECT *
FROM STAFF
WHERE STAFF_ROLE = 'DOCTOR';
```

1SELECT *
2FROM STAFF
3WHERE STAFF_ROLE = 'DOCTOR';
4

Results

Explain

Describe

Saved SQL

History

7

ZULAIKHA BINTI AHMAD

194627890

NO 136, JALAN BUKIT LAMA, MELAKA

ikha18

abc123456789

DOCTOR

01/18/1996

01/09/2025

3

2

29

SUPERVISEE

4

BAKAR BIN SUFIAN

119822727

67 JALAN JASIN, 77000 MELAKA, MALAYSIA

bakarali

abc123456789

DOCTOR

09/16/1982

01/21/2020

2

2

42

SUPERVISEE

3

AMIRAH NUR ALI

138955031

88 TAMAN BUKIT BERUANG, 75450 MELAKA, MALAYSIA

amirahhanan

abc123456789

DOCTOR

01/12/1984

01/11/2024

1

-

41

SUPERVISOR

2

RACHEL KIM ALLISON

143219888

123 JALAN MELAKA RAYA 12, 75000 MELAKA, MALAYSIA

kimallison123

abc123456789

DOCTOR

01/10/1969

08/25/2010

2

-

56

SUPERVISOR

11

NUR SYAKILLA HASAN

0194627334

NO 136, JALAN BUKIT PILAH, MELAKA

syakilla0302

syakillahasan

DOCTOR

01/01/1974

01/01/2000

1

5

40

-

3. RETRIEVE DATA ADMIN (JOIN)

```
SELECT A.STAFF_ID, S.STAFF_NAME, A.ADMIN_START_DATE
FROM STAFF S
JOIN ADMIN A
ON A.STAFF_ID = S.STAFF_ID
```

```
1 SELECT A.STAFF_ID, S.STAFF_NAME, A.ADMIN_START_DATE
2 FROM STAFF S
3 JOIN ADMIN A
4 ON A.STAFF_ID = S.STAFF_ID
```

STAFF_ID	STAFF_NAME	ADMIN_START_DATE
T31	KAMIL	01/17/2025
T32	MARIA	02/11/2022

2 rows returned in 0.00 seconds [Download](#)

4. DISPLAY STAFF THAT HAS NO APPOINTMENT (SUBQUERY)

```
SELECT STAFF_ID, STAFF_NAME
FROM STAFF
WHERE STAFF_ID IN (
    SELECT STAFF_ID
    FROM DOCTOR)
AND STAFF_ID NOT IN (
    SELECT DISTINCT STAFF_ID
    FROM APPOINTMENT);
```

```
1 SELECT STAFF_ID, STAFF_NAME
2 FROM STAFF
3 WHERE STAFF_ID IN (
4     SELECT STAFF_ID
5     FROM DOCTOR)
6 AND STAFF_ID NOT IN (
7     SELECT DISTINCT STAFF_ID
8     FROM APPOINTMENT);
9
10
```

STAFF_ID	STAFF_NAME
189	DANIEL SAHID 2
169	FIFI ABDULLAH
170	Arep Aziz

3 rows returned in 0.01 seconds [Download](#)

5. FIND THE OLDEST STAFF (ORDER BY)

```
SELECT STAFF_ID, STAFF_NAME, STAFF_DOB
FROM STAFF
ORDER BY STAFF_DOB ASC;
```

```
1 SELECT STAFF_ID, STAFF_NAME, STAFF_DOB
2 FROM STAFF
3 ORDER BY STAFF_DOB ASC
4
```

STAFF_ID	STAFF_NAME	STAFF_DOB
170	Arep Aziz	01/29/1973
169	FIFI ABDULLAH	01/18/1988
131	KAMIL	06/24/1991
136	NOOR AMIRAH	04/20/1992
134	Haziq Aqmal	03/12/1993
130	MUHAMMAD	12/01/1994
128	Aqil	01/01/1995
188	MUNIRAH ABDULLAH	01/10/1995
182	DANIEL AKMAL	01/18/1995
192	ADD NURSE	01/16/1996

6. FIND EARLIEST APPOINTMENT DATE

```
SELECT PATIENT_ID, MIN(APPOINTMENT_TIME) AS "EARLIEST
APPOINTMENT"
FROM APPOINTMENT
WHERE PATIENT_ID = '8787'
GROUP BY PATIENT_ID;
```

```
1 SELECT PATIENT_ID, MIN(APPOINTMENT_TIME) AS "EARLIEST APPOINTMENT"
2 FROM APPOINTMENT
3 WHERE PATIENT_ID = '8787'
4 GROUP BY PATIENT_ID;
5
```

PATIENT_ID	EARLIEST APPOINTMENT
8787	16:58

1 rows returned in 0.00 seconds [Download](#)

7. RETRIEVE DETAILS APPOINTMENT BY ID

```
SELECT A.APPOINTMENT_ID, A.APPOINTMENT_DATE,  
A.APPOINTMENT_TIME, S.STAFF_NAME AS Doctor, P.PATIENT_NAME AS  
Patient  
FROM APPOINTMENT A  
JOIN STAFF S  
ON A.STAFF_ID = S.STAFF_ID  
JOIN PATIENT P  
ON A.PATIENT_ID = P.PATIENT_ID  
WHERE A.APPOINTMENT_ID = 37;
```

```
1 SELECT A.APPOINTMENT_ID, A.APPOINTMENT_DATE, A.APPOINTMENT_TIME, S.STAFF_NAME AS Doctor, P.PATIENT_NAME AS Patient
2 FROM APPOINTMENT A
3 JOIN STAFF S ON A.STAFF_ID = S.STAFF_ID
4 JOIN PATIENT P ON A.PATIENT_ID = P.PATIENT_ID
5 WHERE A.APPOINTMENT_ID = 37;
```

APPOINTMENT_ID	APPOINTMENT_DATE	APPOINTMENT_TIME	DOCTOR	PATIENT
37	01/10/2025	16:58	MUHAMMAD	Milia

1 rows returned in 0.00 seconds [Download](#)

8. FIND THE NAMES OF ALL NURSES WHO HAVE A CERTIFICATE LEVEL HIGHER THAN 'LICENSED PRACTICAL NURSE'.

```
SELECT S.STAFF_ID, S.STAFF_NAME  
FROM NURSE N  
JOIN STAFF S ON N.STAFF_ID = S.STAFF_ID  
WHERE N.CERTIFICATE_LEVEL > 'Licensed Practical Nurse';
```

```
1 SELECT s.STAFF_ID, s.STAFF_NAME
2 FROM NURSE n
3 JOIN STAFF s ON n.STAFF_ID = s.STAFF_ID
4 WHERE n.CERTIFICATE_LEVEL > 'Licensed Practical Nurse';
```

STAFF_ID	STAFF_NAME
136	NOOR AMIRAH
134	Haziq Aqmal

2 rows returned in 0.00 seconds [Download](#)

9. FIND THE NAMES OF ALL PATIENTS WHO HAVE HAD MORE THAN TWO APPOINTMENTS WITH A SPECIFIC DOCTOR.

```
SELECT DISTINCT P.PATIENT_NAME
FROM PATIENT P
JOIN APPOINTMENT A ON P.PATIENT_ID = A.PATIENT_ID
JOIN STAFF S ON A.STAFF_ID = S.STAFF_ID
WHERE S.STAFF_ID = 130;
```

```
1 SELECT DISTINCT p.PATIENT_NAME
2 FROM PATIENT p
3 JOIN APPOINTMENT a ON p.PATIENT_ID = a.PATIENT_ID
4 JOIN STAFF s ON a.STAFF_ID = s.STAFF_ID
5 WHERE s.STAFF_ID = 130;
```

PATIENT_NAME
Aliyah
Haziq
Milia
Aqil

4 rows returned in 0.00 seconds [Download](#)

10. COUNT TOTAL STAFF

```
SELECT D.DEPARTMENT_NAME, COUNT(S.STAFF_ID) AS STAFF_COUNT
FROM DEPARTMENT D
LEFT JOIN STAFF S ON D.DEPARTMENT_ID = S.DEPARTMENT_ID
GROUP BY D.DEPARTMENT_NAME;
```

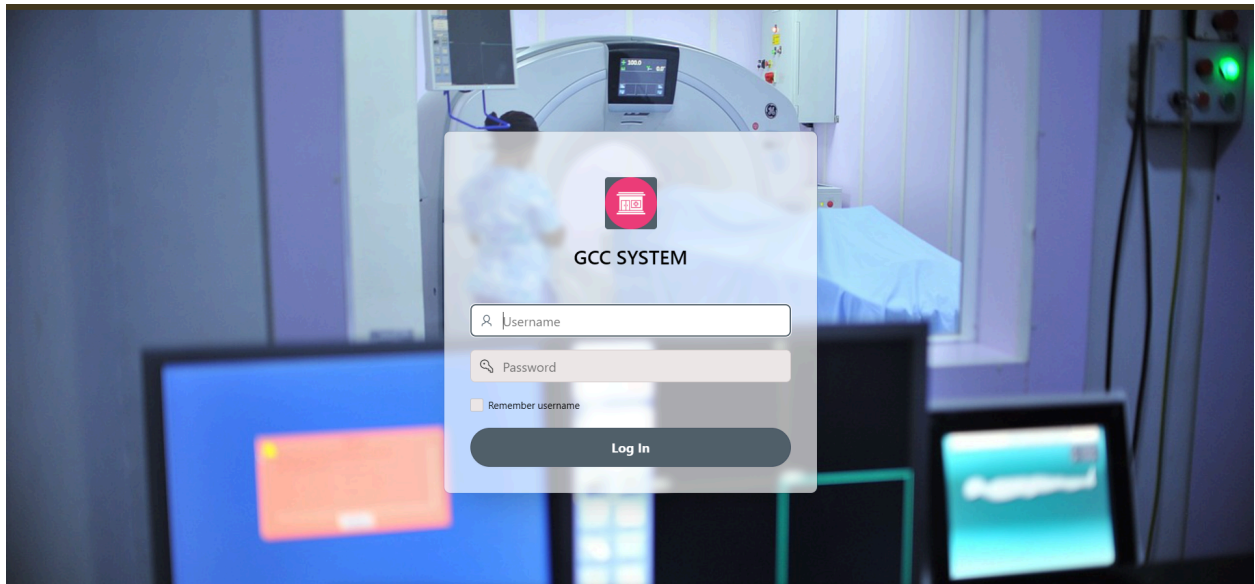
```
1 SELECT D.DEPARTMENT_NAME, COUNT(S.STAFF_ID) AS STAFF_COUNT
2 FROM DEPARTMENT D
3 LEFT JOIN STAFF S ON D.DEPARTMENT_ID = S.DEPARTMENT_ID
4 GROUP BY D.DEPARTMENT_NAME;
```

DEPARTMENT_NAME	STAFF_COUNT
Orthodontist	2
Neurologist	4
Urologist	4
Dermatologist	7
Pharmacist	3
Test2	1

6 rows returned in 0.01 seconds [Download](#)

6.0 SYSTEM DEVELOPMENT SAMPLE SCREEN

USER LOGIN (ADMIN, DOCTOR AND NURSE)



DASHBOARD (ADMIN, DOCTOR AND NURSE)

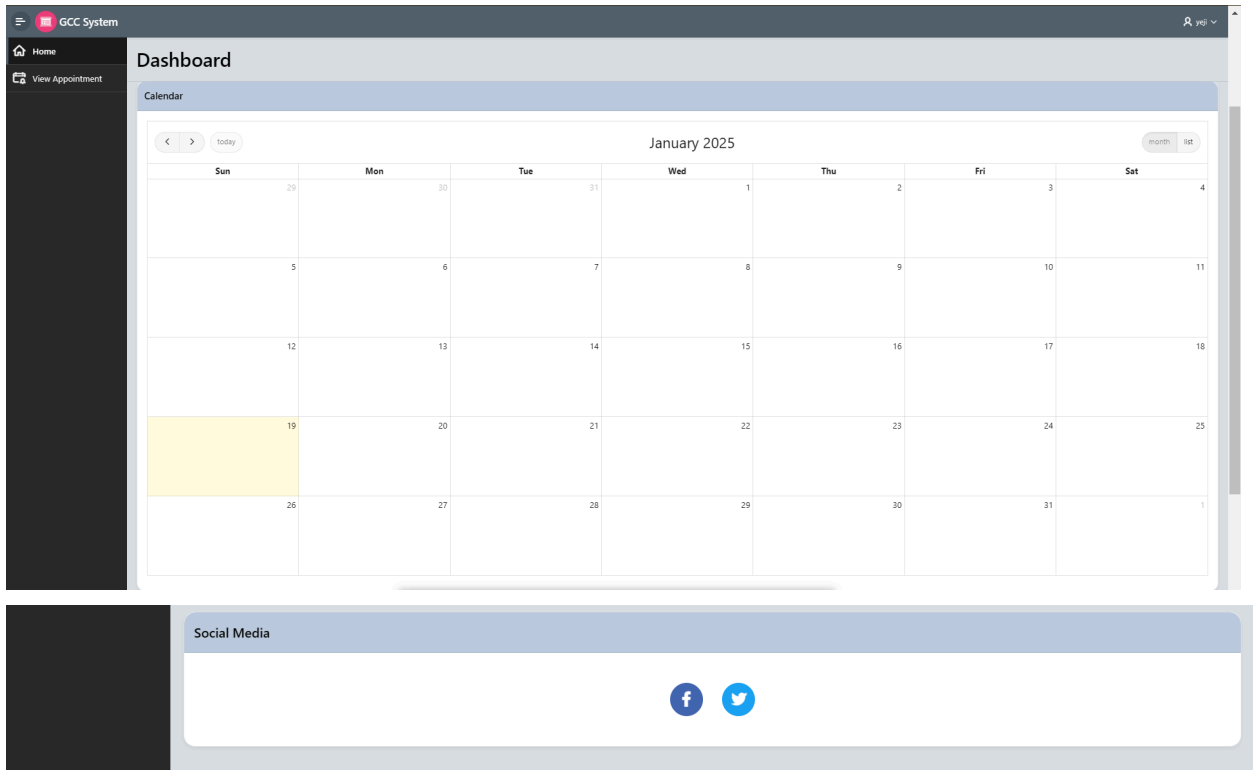
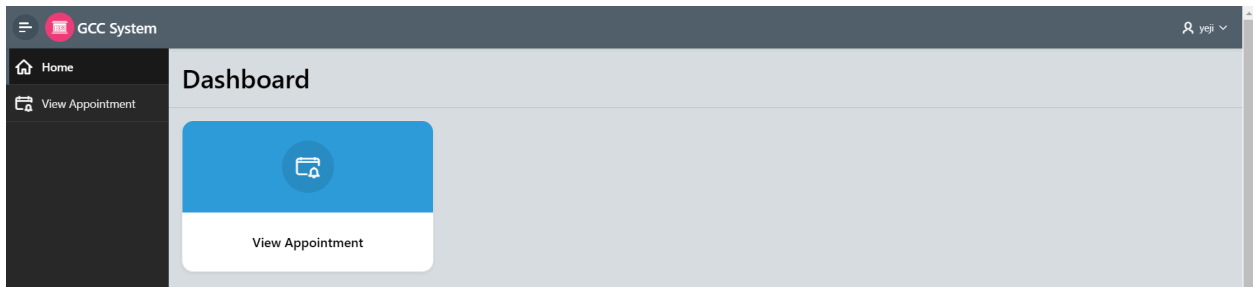
ADMIN DASHBOARD

The screenshot displays the Admin Dashboard of the GCC System. The interface includes a dark sidebar with navigation links: Home, Manage Staff, Manage Department, and Manage Appoin... (Appointment). The main content area is titled "Dashboard" and features two large buttons: "Manage Staff" and "Manage Department". Below these is a "Calendar" section showing a monthly view for January 2025. The calendar grid shows dates from 1 to 18, with the 19th highlighted in yellow. At the bottom of the dashboard is a "Social Media" section with icons for Facebook and Twitter.

Calendar Data (January 2025):

Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1

DOCTOR DASHBOARD



NURSE DASHBOARD

 GCC System

 sitaib ▾

Dashboard


Manage Appointment


Book Appointment


Manage Patient

Calendar

<

>

today

January 2025

month

list

Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1

Social Media



READ (ADMIN, DOCTOR AND NURSE)

ADMIN

READ data from Manage Staff

GCC System

admin

Manage Staff

Go

Actions

Add New Staff

	Name	Phone Number	Address	System Username	Position	Level	DOB	Age	Department	Supervisor	Hire Date	System Password
	SITI BINTI TALIBAH	123111233	2 JALAN MERSING, 76100 MELAKA, MALAYSIA.	sitalib	NURSE	SUPERVISOR	1/1/1976	49	Pharmacist		2/15/2024	abc123456789
	ZULAIKHA BINTI AHMAD	194627890	NO 136,JALAN BUKIT LAMA, MELAKA	ikha18	DOCTOR	SUPERVISEE	1/18/1996	29	Neurologist	RACHEL KIM ALLISON	1/9/2025	abc123456789
	BAKAR BIN SUFIAN	119822727	67 JALAN JASIN, 77000 MELAKA, MALAYSIA	bakarali	DOCTOR	SUPERVISEE	9/16/1982	42	Pharmacist	RACHEL KIM ALLISON	1/21/2020	abc123456789
	AMIRAH NUR ALI	138955031	88 TAMAN BUKIT BERJANG, 75450 MELAKA, MALAYSIA	amirahhanan	DOCTOR	SUPERVISOR	1/12/1984	41	Dermatologist		1/17/2024	abc123456789
	FAZILAH BINTI ABDULLAH	138922131	111 TAMAN BUKIT RAYA, 75450 MELAKA, MALAYSIA	fazilah123	NURSE	SUPERVISEE	1/8/1998	27	Pharmacist	SITI BINTI TALIBAH	1/1/2025	abc123456789
	RACHEL KIM ALLISON	143219888	123 JALAN MELAKA RAYA 12, 75000 MELAKA, MALAYSIA	kimallison123	DOCTOR	SUPERVISOR	1/10/1969	56	Pharmacist		8/25/2010	abc123456789
	NUR SYAKILLA HASAN	0194627334	NO 136, JALAN BUKIT PILAH,MELAKA	syakilla0302	DOCTOR		1/1/1974	40	Dermatologist	SITI BINTI TALIBAH	1/1/2000	syakillahasan
	YEJI	138955601	NO 20 TAMAN INDAH, MELAKA	Yeji	DOCTOR	SUPERVISOR	1/15/2010	15	Dermatologist		1/21/2025	Yeji 123

1 - 8

READ data from Doctor

GCC System

admin

Doctor

Go

Actions

Add New Doctor

Add New Nurse

Doctor Name	Specialization	Room Number	Medical License Number
BAKAR BIN SUFIAN	Pharmacy Research	A2	P23212
RACHEL KIM ALLISON	Pharmaceutical Care	A2	P12345
AMIRAH NUR ALI	Skin dermatology	A3	M12345
ZULAIKHA BINTI AHMAD	Skin	A7	M12367

1 - 4

READ data from Nurse

GCC System

admin

Nurse

Go

Actions

Add New Nurse

Nurse Name	Certificate Level
SITI BINTI TALIBAH	Bachelor of Science in Nursing
FAZILAH BINTI ABDULLAH	Bachelor of Science in Nursing

1 - 2

READ data from Supervisor

GCC System			admin
Supervisor			
Q Go Actions			
Supervisor ID	Name	Position	
5	SITI BINTI TALIBAH	NURSE	
3	AMIRAH NUR ALI	DOCTOR	
2	RACHEL KIM ALLISON	DOCTOR	
9	YEJI	DOCTOR	
			1 - 4

READ data from Manage Department

GCC System			admin
Manage Department			
Q Go Actions			Add New Department
	Department ID	Department Name	Department Description
	1	Dermatologist	Dermatologist
	2	Pharmacist	Pharmacist
	3	Neurologist	Neuron
	4	Urologist	Urologist
	5	Orthodontist	Orthodontist
			1 - 5

READ data from Manage Appointment

GCC System			admin
Manage Appointment			
Q Go Actions			
Appointment Date	Doctor Assigned	Patient Name	Appointment Time
Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	10:00 AM
Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	03:00 PM
Thursday, 09 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	04:00 PM
Friday, 17 January, 2025	BAKAR BIN SUFIAN	Milla Ahmad	03:00 PM
Friday, 17 January, 2025	AMIRAH NUR ALI	Milla Ahmad	03:00 PM
Thursday, 16 January, 2025	NUR SYAKILLA HASAN	Milla Ahmad	03:00 PM
Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	05:00 PM
Tuesday, 07 January, 2025	ZULAIKHA BINTI AHMAD	Milla Ahmad	06:00 PM
Sunday, 19 January, 2025	RACHEL KIM ALLISON	Aliyah Binti Ali	10:00 AM
			1 - 9

DOCTOR

READ data from View Appointment

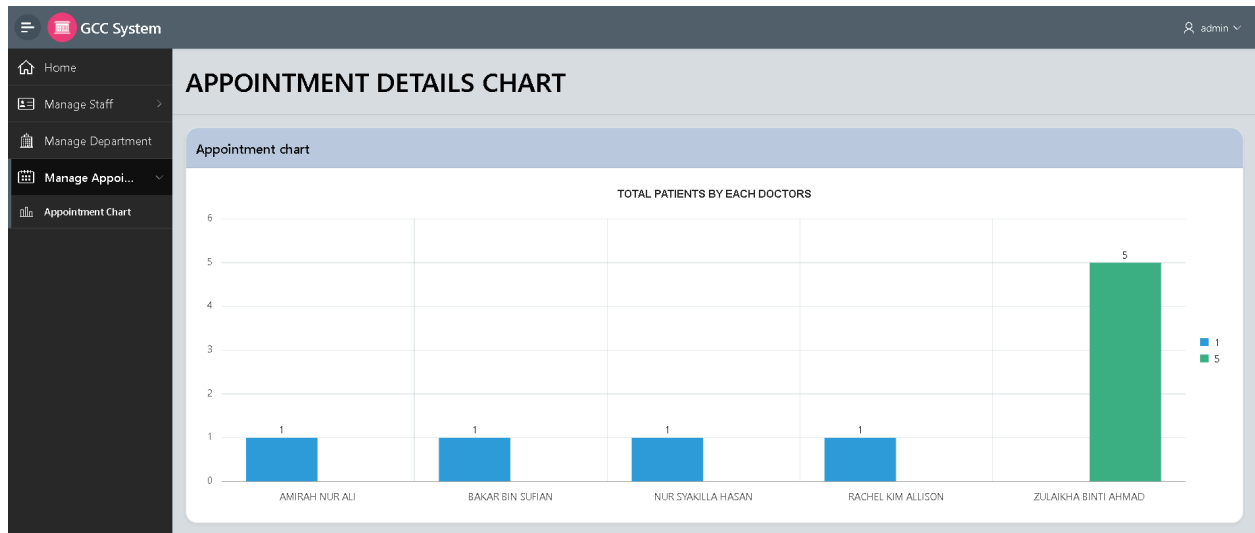
GCC System				
View Appointment				
Q Go Actions Group By				
Appointment Date	Appointment Time	Doctor	Patient	
1/7/2025	06:00 PM	ZULAIKHA BINTI AHMAD	Milla Ahmad	
1/2/2025	10:00 AM	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	
1/9/2025	04:00 PM	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	
1/16/2025	03:00 PM	NUR SYAKILLA HASAN	Milla Ahmad	
1/2/2025	03:00 PM	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	
1/19/2025	10:00 AM	RACHEL KIM ALLISON	Aliyah Binti Ali	
1/17/2025	03:00 PM	AMIRAH NUR ALI	Milla Ahmad	
1/17/2025	03:00 PM	BAKAR BIN SUFIAN	Milla Ahmad	
1/2/2025	05:00 PM	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	
				1 - 9

NURSE

READ data from Manage Appointment

GCC System				
Manage Appointment				
Q Go Actions Book Appointment				
	Appointment Date	Doctor Assigned	Patient Name	Appointment Time
	Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	10:00 AM
	Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	03:00 PM
	Thursday, 09 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	04:00 PM
	Friday, 17 January, 2025	BAKAR BIN SUFIAN	Milla Ahmad	03:00 PM
	Friday, 17 January, 2025	AMIRAH NUR ALI	Milla Ahmad	03:00 PM
	Thursday, 16 January, 2025	NUR SYAKILLA HASAN	Milla Ahmad	03:00 PM
	Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	05:00 PM
	Tuesday, 07 January, 2025	ZULAIKHA BINTI AHMAD	Milla Ahmad	06:00 PM
	Sunday, 19 January, 2025	RACHEL KIM ALLISON	Aliyah Binti Ali	10:00 AM
				1 - 9

READ data from APPOINTMENT DETAILS CHART



INSERT (ADMIN, DOCTOR AND NURSE)

ADMIN

Insert new staff

GCC System admin

Add New Staff

Staff Detail

Auto Generated Staff ID

Staff Name Phone Number

Date of Birth Age

Address

Hire Date Position Doctor Level Supervisee

Department Dermatologist Supervisor

System Username

System Password

Cancel Add Staff

Insert new Doctor

GCC System admin

Add Doctor

Add Doctor

Doctor Name Specialization

Room Number Med License Num

Cancel Add Doctor

Insert Nurse details

Bachelor of Science in Nursing

Add Nurse

Nurse Name

▼

Certificate Level

Cancel

Add Nurse

Insert department details

Department Detail

Department ID

Department Name

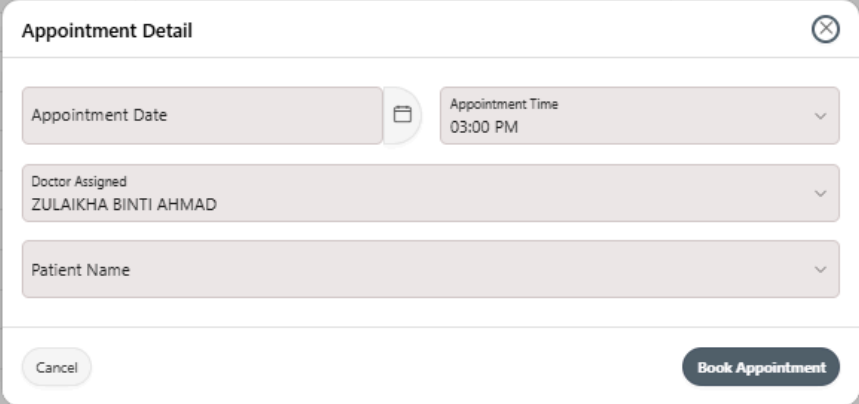
Department Description

Cancel

Add Department

NURSE

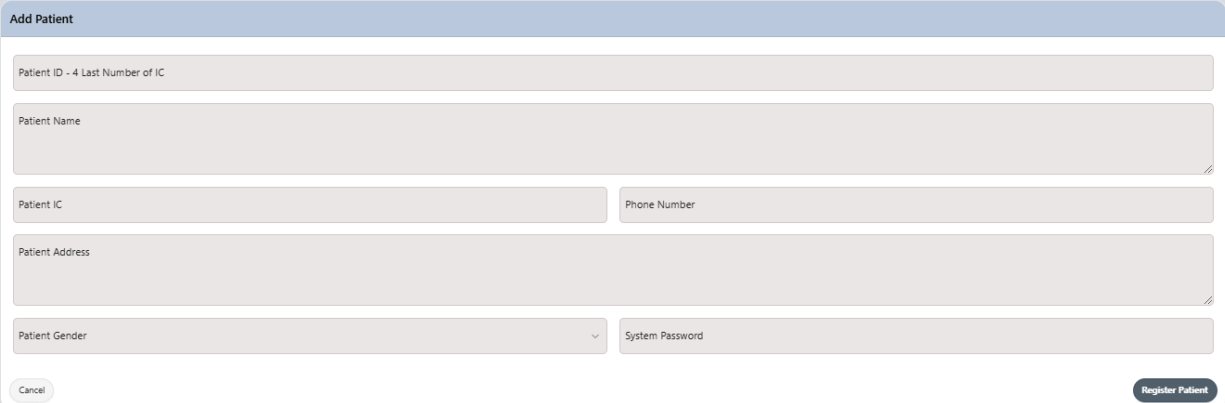
Insert appointment details



The image shows a modal window titled "Appointment Detail" overlaid on a table. The table has columns for "Appointment Date", "Doctor Assigned", "Patient Name", and "Appointment Time". The modal contains the following fields:

- Appointment Date:** A text input field with a calendar icon.
- Appointment Time:** A dropdown menu showing "03:00 PM".
- Doctor Assigned:** A dropdown menu showing "ZULAIKHA BINTI AHMAD".
- Patient Name:** A dropdown menu.
- Buttons:** "Cancel" and "Book Appointment".

Insert new patient



The image shows a form titled "Add Patient" within a system interface. The form includes the following fields:

- Patient ID - 4 Last Number of IC:** A text input field.
- Patient Name:** A text input field.
- Patient IC:** A text input field.
- Phone Number:** A text input field.
- Patient Address:** A text input field.
- Patient Gender:** A dropdown menu.
- System Password:** A text input field.
- Buttons:** "Cancel" and "Register Patient".

UPDATE & DELETE (ADMIN AND NURSE)

ADMIN

DELETE and UPDATE for staff details

Staff Detail

Name

ZULAIKHA BINTI AHMAD

Phone Number

194627890

Date of Birth

1/18/1996

Age

29

Staff Address

NO 136,JALAN BUKIT LAMA, MELAKA

Hire Date

1/9/2025

Position

Doctor

Level

Supervisee

Department

Neurologist

Supervisor

RACHEL KIM ALLISON

System Username

ikha18

System Password

*****789

Cancel

Delete

Apply Changes

DELETE and UPDATE for department details

Department Detail

Department ID

1

Department Name

Dermatologist

Department Description

Dermatologist

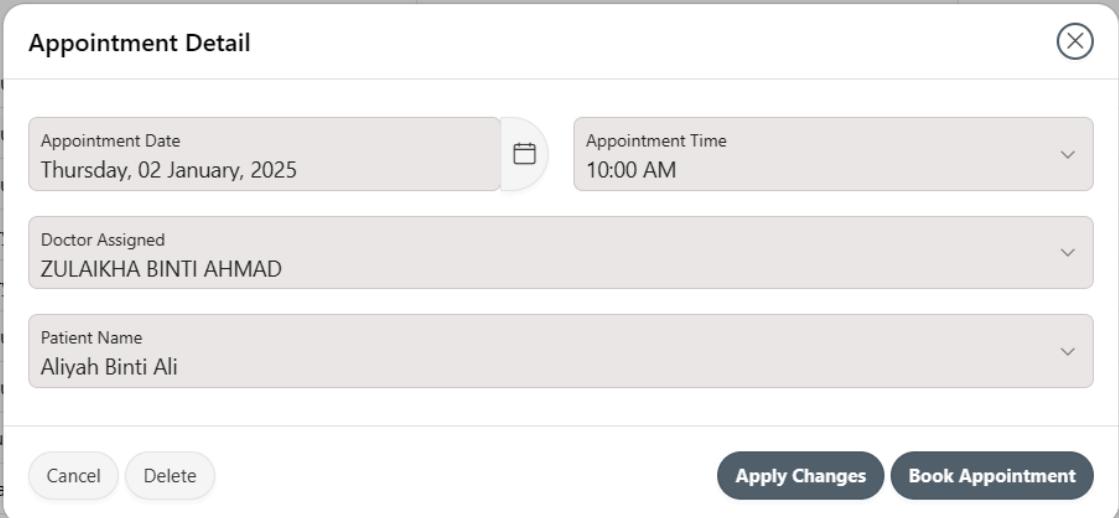
Cancel

Delete

Apply Changes

NURSE

DELETE and UPDATE for appointment details



The image shows a web application interface with a modal window titled "Appointment Detail". The modal contains four input fields: "Appointment Date" with the value "Thursday, 02 January, 2025" and a calendar icon; "Appointment Time" with the value "10:00 AM" and a dropdown arrow; "Doctor Assigned" with the value "ZULAIKHA BINTI AHMAD" and a dropdown arrow; and "Patient Name" with the value "Aliyah Binti Ali" and a dropdown arrow. At the bottom of the modal are four buttons: "Cancel", "Delete", "Apply Changes", and "Book Appointment". The background shows a calendar grid with dates from January 02 to January 09.

Go Actions ▾

Appointment Detail ✕

Appointment Date Thursday, 02 January, 2025 📅

Appointment Time 10:00 AM ▾

Doctor Assigned ZULAIKHA BINTI AHMAD ▾

Patient Name Aliyah Binti Ali ▾

Cancel Delete Apply Changes Book Appointment

DELETE and UPDATE for patient details

tail

Patient

Patient ID

153

Patient Name

Aliyah Binti Ali

Patient IC

010181-04-0153

Phone Number

131100928

Patient Address

43, Jalan Kulai, Kedah

Patient Gender

Female

Patient Password

*****nia

Cancel

Delete

Apply Changes

23

BRIDGE

GCC System

admin

Home

Manage Staff

Manage Department

Manage Appoin...

Appointment Chart

Manage Appointment

Q

Go

Actions

Appointment Date	Doctor Assigned	Patient Name	Appointment Time
Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	10:00 AM
Thursday, 09 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	04:00 PM
Friday, 17 January, 2025	BAKAR BIN SUFIAN	Milla Ahmad	03:00 PM
Thursday, 16 January, 2025	NUR SYAKILLA HASAN	Milla Ahmad	03:00 PM
Thursday, 02 January, 2025	ZULAIKHA BINTI AHMAD	Aliyah Binti Ali	05:00 PM
Tuesday, 07 January, 2025	ZULAIKHA BINTI AHMAD	Milla Ahmad	06:00 PM
Sunday, 19 January, 2025	RACHEL KIM ALLISON	Aliyah Binti Ali	10:00 AM

1 - 7

GCC System

admin

Home

Manage Staff

Manage Department

Manage Appoi...

Appointment Chart

APPOINTMENT DETAILS CHART

Appointment chart

TOTAL PATIENTS BY EACH DOCTORS

Doctor	Total Patients
AMIRAH NUR ALI	1
BAKAR BIN SUFIAN	1
NUR SYAKILLA HASAN	1
RACHEL KIM ALLISON	1
ZULAIKHA BINTI AHMAD	5

RECURSIVE

GCC System

admin

Home

Manage Staff

Doctor

Nurse

Supervisor

Manage Department

Manage Appoint...

Supervisor

Q

Go

Actions

Supervisor ID	Name	Position
5	SITI BINTI TALIBAH	NURSE
3	AMIRAH NUR ALI	DOCTOR
2	RACHEL KIM ALLISON	DOCTOR
9	YEJI	DOCTOR

1 - 4

INHERITANCE

GCC System

Home

Manage Staff

Doctor

Nurse

Supervisor

Manage Department

Manage Appoint...

Manage Department

Q

Go

Actions

	Department ID	Dep
	1	Der
	2	Pha
	3	Neu
	4	Uro
	5	Ortl

EXTRA

i. Password Mask

Patient Detail						
Go Actions + Add Patient						
	Name ↑	Identification Number	Phone Number	Address	Gender	System Password
☐	Aliyah Binti Ali	010181-04-0153	131100928	43, Jalan Kulai, Kedah	Female	*****nia
☐	Milla Ahmad	010510-02-0101	122123453	45 Jalan Tun Ali, 75300 Melaka, Malaysia	Female	****abc
1 - 2						

PATIENT						
+ Model Constraints Grants Statistics UI Defaults Triggers Dependencies SQL REST Sample Queries						
Insert Row Load Data						
	PATIENT_NAME	PATIENT_IC	PATIENT_PNO	PATIENT_ADDRESS	PATIENT_GENDER	PATIENT_PASSWORD
←	Milla Ahmad	010510-02-0101	122123453	45 Jalan Tun Ali, 75300 Melaka, Malaysia	Female	****abc
	Aliyah Binti Ali	010181-04-0153	131100928	43, Jalan Kulai, Kedah	Female	*****nia

ii. Use of triggers for sequence, and password mask

```
1 SELECT *
2 FROM USER_TRIGGERS;
```

DESCRIPTION	ACTION_TYPE	TRIGGER_BODY	CROSSEDITION	BEFORE_STATEMENT	BEFORE_ROW	AFTER_ROW	AFTER_STATEMENT	INSTEAD_OF_ROW	FIRE_ONCE
appointment_id.trigger BEFORE INSERT ON appointment FOR EACH ROW	PL/SQL	BEGIN -- Assign the next value of the sequence to the staff_id column :NEW.appointment_id := appointment_id_seq.NEXTVAL; END;	NO	NO	NO	NO	NO	NO	YES
staff_id.trigger BEFORE INSERT ON staff FOR EACH ROW	PL/SQL	BEGIN :NEW.staff_id := staff_id_seq.NEXTVAL; END;	NO	NO	NO	NO	NO	NO	YES
trg_department_id BEFORE INSERT ON department FOR EACH ROW	PL/SQL	BEGIN :NEW.department_id := department_id_seq.NEXTVAL; END;	NO	NO	NO	NO	NO	NO	YES
TRG_MASK_PATIENT_PASSWORD BEFORE INSERT OR UPDATE ON PATIENT FOR EACH ROW	PL/SQL	BEGIN -- Ensure the password is masked before storing IF :NEW.PATIENT_PASSWORD IS NOT NULL THEN :NEW.PATIENT_PASSWORD := RPAD('*', LENGTH(:NEW.PATIENT_PASSWORD) - 3, '*') SUBSTR(:NEW.PATIENT_PASSWORD, -3); END IF; END;	NO	NO	NO	NO	NO	NO	YES

```
SELECT SEQUENCE_NAME, LAST_NUMBER, MIN_VALUE, MAX_VALUE, INCREMENT_BY, CYCLE_FLAG, CACHE_SIZE
FROM USER_SEQUENCES;
```

7.0 CONCLUSION

GCC booking and appointment system represents a transformative step forward in improving healthcare management. It aims to overcome the drawbacks of manual and file-based systems by addressing the pain points of booking appointments, record-keeping, and operational efficiency. The system covers most of the essential features required to streamline clinic operations. Different user roles, including Doctor, Nurse, and Admin, were successfully implemented, with specific access and functionality tailored to each role.

Doctors can view appointments, although filtering to display only appointments associated with each doctor has not been fully implemented. Nurses are able to view all appointments and manage patient details. However, there are unresolved issues with booking and editing appointments, such as preventing time conflicts and ensuring that doctor details remain visible after updates. Admins can manage staff, department, and appointment details effectively, but editing additional information specific to doctors or nurses, such as certifications or specializations, is not currently supported.

Although the system successfully addresses many key functionalities, certain features remain incomplete due to skill and time constraints. These include filtering appointments for doctors, implementing validation to prevent overlapping appointment times, and enabling more detailed data editing for admin users. These areas present opportunities for further development to refine the system.

Overall, the system still establishes a strong foundation for managing clinic operations by offering a user-friendly interface, enabling clinic staff to store information about appointments seamlessly. While there are some challenges and limitations, the current implementation provides significant functionality. Future enhancements, such as better data filtering, improved time validation, and expanded admin tools, can further enhance the system's performance and usability.

8.0 APPENDIX

8.1 ENTITY RELATIONSHIP DIAGRAM

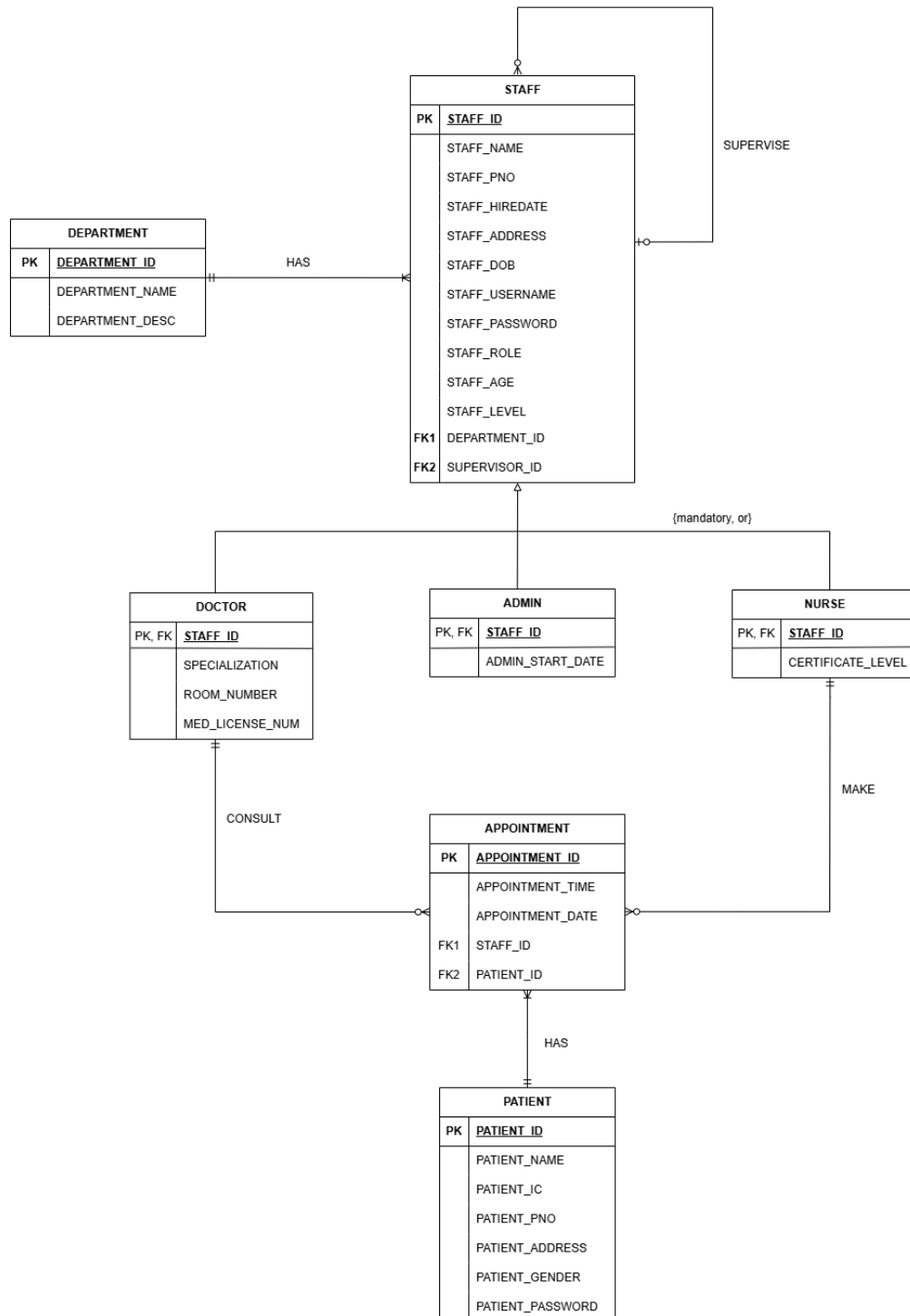


Figure 8.1: Entity Relationship Diagram (ERD)

8.2 3NF RELATIONAL SCHEMA

DEPARTMENT (DEPARTMENT_ID, DEPARTMENT_NAME,
DEPARTMENT_DESC)

STAFF (STAFF_ID, STAFF_NAME, STAFF_PNO, STAFF_ADDRESS,
STAFF_USERNAME, STAFF_PASSWORD, STAFF_ROLE,
STAFF_HIREDATE, STAFF_DOB, STAFF_AGE, STAFF_LEVEL
DEPARTMENT_ID*, SUPERVISOR_ID*)

DOCTOR (STAFF_ID*, SPECIALIZATION, ROOM_NUMBER,
MED_LICENSE_NUM)

NURSE (STAFF_ID*, CERTIFICATE_LEVEL)

ADMIN (STAFF_ID*, ADMIN_START_DATE)

PATIENT (PATIENT_ID, PATIENT_NAME, PATIENT_IC, PATIENT_PNO,
PATIENT_ADDRESS, PATIENT_GENDER, PATIENT_PASSWORD)

APPOINTMENT (APPOINTMENT_ID, APPOINTMENT_TIME,
APPOINTMENT_DATE, STAFF_ID*, PATIENT_ID*)

8.3 DATA DICTIONARY

TABLE NAME	ATTRIBUTE NAME	DATA TYPE	PK or FK	REFERENCE TABLE
DEPARTMENT	DEPARTMENT_ID	NUMBER (10)	PK	
	DEPARTMENT_NAME	VARCHAR2 (20)		
	DEPARTMENT_DESC	VARCHAR2 (200)		
STAFF	STAFF_ID	NUMBER (4)	PK	
	STAFF_NAME	VARCHAR2 (100)		
	STAFF_PNO	VARCHAR2 (15)		
	STAFF_ADDRESS	VARCHAR2 (200)		
	STAFF_USERNAME	VARCHAR2 (10)		
	STAFF_PASSWORD	VARCHAR2 (20)		
	STAFF_ROLE	VARCHAR2 (10)		
	STAFF_DOB	DATE		
	STAFF_HIREDATE	DATE		
	STAFF_AGE	NUMBER (3)		
	STAFF_LEVEL	VARCHAR2 (50)		
	DEPARTMENT_ID	NUMBER (10)	FK	DEPARTMENT
	SUPERVISOR_ID	NUMBER (4)	FK	STAFF
ADMIN	STAFF_ID	NUMBER (4)	PK, FK	STAFF
	ADMIN_START_DATE	DATE		
DOCTOR	STAFF_ID	NUMBER (4)	PF, FK	STAFF
	SPECIALIZATION	VARCHAR (100)		

	ROOM_NUMBER	VARCHAR2 (10)		
	MED_LICENSE_NUM	VARCHAR2 (20)		
NURSE	STAFF_ID	NUMBER (4)	PK, FK	STAFF
	CERTIFICATE_LEVEL	VARCHAR2 (30)		
PATIENT	PATIENT_ID	NUMBER (12)	PK	
	PATIENT_NAME	VARCHAR2 (100)		
	PATIENT_IC	VARCHAR2 (12)		
	PATIENT_PNO	VARCHAR2 (15)		
	PATIENT_ADDRESS	VARCHAR2 (200)		
	PATIENT_GENDER	VARCHAR2 (10)		
	PATIENT_PASSWORD	VARCHAR2 (20)		
APPOINTMENT	APPOINTMENT_ID	NUMBER (4)	PK	
	APPOINTMENT_TIME	DATE		
	APPOINTMENT_DATE	DATE		
	STAFF_ID	NUMBER (4)	FK	STAFF
	PATIENT_ID	NUMBER (12)	FK	PATIENT

8.4 DATA DEFINITION LANGUAGE (DDL)

8.4.1. Table Department

```
CREATE TABLE "DEPARTMENT"
(
  "DEPARTMENT_ID" NUMBER(10,0),
  "DEPARTMENT_NAME" VARCHAR2(20) COLLATE "USING_NLS_COMP",
  "DEPARTMENT_DESC" VARCHAR2(200) COLLATE "USING_NLS_COMP",
  CONSTRAINT "DEPARTMENT_ID_NN" CHECK (department_id IS NOT NULL) ENABLE,
  CONSTRAINT "DEPARTMENT_NAME_NN" CHECK (department_name IS NOT NULL) ENABLE,
  CONSTRAINT "DEPARTMENT_DESC_NN" CHECK (department_desc IS NOT NULL) ENABLE,
  CONSTRAINT "DEPARTMENT_ID_PK" PRIMARY KEY ("DEPARTMENT_ID")
  USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/

CREATE OR REPLACE EDITIONABLE TRIGGER "TRG_DEPARTMENT_ID"
BEFORE INSERT ON department
FOR EACH ROW
BEGIN
  :NEW.department_id := department_id_seq.NEXTVAL;
END;

/
ALTER TRIGGER "TRG_DEPARTMENT_ID" ENABLE
/
```

Column Name	Data Type	Nullable	Default	Primary Key
DEPARTMENT_ID	NUMBER(10,0)	No	-	1
DEPARTMENT_NAME	VARCHAR2(20)	Yes	-	-
DEPARTMENT_DESC	VARCHAR2(200)	Yes	-	-

Figure 8.4.1 Table department

8.4.2. Table Staff

```
CREATE TABLE "STAFF"
(
    "STAFF_ID" NUMBER(4,0),
    "STAFF_NAME" VARCHAR2(100) COLLATE "USING_NLS_COMP",
    "STAFF_PNO" VARCHAR2(15) COLLATE "USING_NLS_COMP",
    "STAFF_ADDRESS" VARCHAR2(200) COLLATE "USING_NLS_COMP",
    "STAFF_USERNAME" VARCHAR2(30) COLLATE "USING_NLS_COMP",
    "STAFF_PASSWORD" VARCHAR2(30) COLLATE "USING_NLS_COMP",
    "STAFF_ROLE" VARCHAR2(10) COLLATE "USING_NLS_COMP",
    "STAFF_DOB" DATE,
    "STAFF_HIREDATE" DATE,
    "DEPARTMENT_ID" NUMBER(10,0),
    "SUPERVISOR_ID" NUMBER(4,0),
    "STAFF_AGE" NUMBER(3,0),
    "STAFF_LEVEL" VARCHAR2(50) COLLATE "USING_NLS_COMP",
    CONSTRAINT "STAFF_ID_NN" CHECK (staff_id IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_NAME_NN" CHECK (staff_name IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_PNO_NN" CHECK (staff_pno IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_ADDRESS_NN" CHECK (staff_address IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_USERNAME_NN" CHECK (staff_username IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_PASSWORD_NN" CHECK (staff_password IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_ROLE_NN" CHECK (staff_role IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_DOB_NN" CHECK (staff_dob IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_HIREDATE_NN" CHECK (staff_hiredate IS NOT NULL) ENABLE,
    CONSTRAINT "STAFF_ID_PK" PRIMARY KEY ("STAFF_ID")
    USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/
ALTER TABLE "STAFF" ADD CONSTRAINT "DEPARTMENT_ID_FK" FOREIGN KEY ("DEPARTMENT_ID")
    REFERENCES "DEPARTMENT" ("DEPARTMENT_ID") ENABLE
/
ALTER TABLE "STAFF" ADD CONSTRAINT "SUPERVISOR_ID_FK" FOREIGN KEY ("SUPERVISOR_ID")
    REFERENCES "STAFF" ("STAFF_ID") ENABLE
/
```

```
CREATE OR REPLACE EDITIONABLE TRIGGER "STAFF_ID_TRIGGER"
BEFORE INSERT ON staff
FOR EACH ROW
BEGIN
    :NEW.staff_id := staff_id_seq.NEXTVAL;
END;
/
ALTER TRIGGER "STAFF_ID_TRIGGER" ENABLE
/

CREATE OR REPLACE EDITIONABLE TRIGGER "TRG_MASK_PASSWORD"
BEFORE INSERT OR UPDATE ON STAFF
FOR EACH ROW
BEGIN
    -- Only mask the password if it is not already masked
    IF :NEW.STAFF_PASSWORD IS NOT NULL THEN
        :NEW.STAFF_PASSWORD := RPAD('*', LENGTH(:NEW.STAFF_PASSWORD) - 3, '*') ||
            SUBSTR(:NEW.STAFF_PASSWORD, -3);
    END IF;
END;
/
ALTER TRIGGER "TRG_MASK_PASSWORD" ENABLE
/
```

Column Name	Data Type	Nullable	Default	Primary Key
STAFF_ID	NUMBER(4,0)	No	-	1
STAFF_NAME	VARCHAR2(100)	Yes	-	-
STAFF_PNO	VARCHAR2(15)	Yes	-	-
STAFF_ADDRESS	VARCHAR2(200)	Yes	-	-
STAFF_USERNAME	VARCHAR2(30)	Yes	-	-
STAFF_PASSWORD	VARCHAR2(30)	Yes	-	-
STAFF_ROLE	VARCHAR2(10)	Yes	-	-
STAFF_DOB	DATE	Yes	-	-
STAFF_HIREDATE	DATE	Yes	-	-
DEPARTMENT_ID	NUMBER(10,0)	Yes	-	-
SUPERVISOR_ID	NUMBER(4,0)	Yes	-	-
STAFF_AGE	NUMBER(3,0)	Yes	-	-

Figure 8.4.2 Table Staff

8.4.3. Table Doctor

```

CREATE TABLE "DOCTOR"
(
  "STAFF_ID" NUMBER(4,0),
  "SPECIALIZATION" VARCHAR2(100) COLLATE "USING_NLS_COMP",
  "ROOM_NUMBER" VARCHAR2(10) COLLATE "USING_NLS_COMP",
  "MED_LICENSE_NUM" VARCHAR2(20) COLLATE "USING_NLS_COMP",
  CONSTRAINT "DOCTOR_STAFF_ID_NN" CHECK (staff_id IS NOT NULL) ENABLE,
  CONSTRAINT "SPECIALIZATION_NN" CHECK (specialization IS NOT NULL) ENABLE,
  CONSTRAINT "ROOM_NUMBER_NN" CHECK (room_number IS NOT NULL) ENABLE,
  CONSTRAINT "MED_LICENSE_NUM_NN" CHECK (med_license_num IS NOT NULL) ENABLE,
  CONSTRAINT "DOCTOR_STAFF_ID_PK" PRIMARY KEY ("STAFF_ID")
  USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/
ALTER TABLE "DOCTOR" ADD CONSTRAINT "DOCTOR_STAFF_ID_FK" FOREIGN KEY ("STAFF_ID")
REFERENCES "STAFF" ("STAFF_ID") ENABLE
/

```

Column Name	Data Type	Nullable	Default	Primary Key
STAFF_ID	NUMBER(4,0)	No	-	1
SPECIALIZATION	VARCHAR2(100)	Yes	-	-
ROOM_NUMBER	VARCHAR2(10)	Yes	-	-
MED_LICENSE_NUM	VARCHAR2(20)	Yes	-	-

Figure 8.4.3 Table Doctor

8.4.4. Table Nurse

```
CREATE TABLE "NURSE"
(
  "STAFF_ID" NUMBER(4,0),
  "CERTIFICATE_LEVEL" VARCHAR2(30) COLLATE "USING_NLS_COMP",
  CONSTRAINT "NURSE_STAFF_ID" CHECK (staff_id IS NOT NULL) ENABLE,
  CONSTRAINT "NURSE_CERTIFICATE_LEVEL_NN" CHECK (certificate_level IS NOT NULL) ENABLE,
  CONSTRAINT "NURSE_STAFF_ID_PK" PRIMARY KEY ("STAFF_ID")
  USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/
ALTER TABLE "NURSE" ADD CONSTRAINT "NURSE_STAFF_ID_FK" FOREIGN KEY ("STAFF_ID")
REFERENCES "STAFF" ("STAFF_ID") ENABLE
/
```

Column Name	Data Type	Nullable	Default	Primary Key
STAFF_ID	NUMBER(4,0)	No	-	1
CERTIFICATE_LEVEL	VARCHAR2(30)	Yes	-	-

Figure 8.4.4 Table Nurse

8.4.5. Table Admin

```
CREATE TABLE "ADMIN"
(
  "STAFF_ID" NUMBER,
  "ADMIN_START_DATE" DATE,
  CONSTRAINT "ADMIN_STAFF_ID_NN" CHECK (staff_id IS NOT NULL) ENABLE,
  CONSTRAINT "ADMIN_START_DATE_NN" CHECK (admin_start_date IS NOT NULL) ENABLE,
  CONSTRAINT "ADMIN_STAFF_ID_PK" PRIMARY KEY ("STAFF_ID")
  USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/
ALTER TABLE "ADMIN" ADD CONSTRAINT "ADMIN_STAFF_ID_FK" FOREIGN KEY ("STAFF_ID")
REFERENCES "STAFF" ("STAFF_ID") ENABLE
/
```

Column Name	Data Type	Nullable	Default	Primary Key
STAFF_ID	NUMBER	No	-	1
ADMIN_START_DATE	DATE	Yes	-	-

Figure 8.4.5 Table Admin

8.4.6. Table Patient

```
CREATE TABLE "PATIENT"
(
  "PATIENT_ID" NUMBER(12,0),
  "PATIENT_NAME" VARCHAR2(100) COLLATE "USING_NLS_COMP",
  "PATIENT_IC" VARCHAR2(15) COLLATE "USING_NLS_COMP",
  "PATIENT_PNO" VARCHAR2(20) COLLATE "USING_NLS_COMP",
  "PATIENT_ADDRESS" VARCHAR2(200) COLLATE "USING_NLS_COMP",
  "PATIENT_GENDER" VARCHAR2(10) COLLATE "USING_NLS_COMP",
  "PATIENT_PASSWORD" VARCHAR2(20) COLLATE "USING_NLS_COMP",
  CONSTRAINT "PATIENT_ID_NN" CHECK (patient_id IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_NAME_NN" CHECK (patient_name IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_IC_NN" CHECK (patient_ic IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_PNO_NN" CHECK (patient_pno IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_ADDRESS_NN" CHECK (patient_address IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_GENDER_NN" CHECK (patient_gender IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_PASSWORD_NN" CHECK (patient_password IS NOT NULL) ENABLE,
  CONSTRAINT "PATIENT_ID_PK" PRIMARY KEY ("PATIENT_ID")
  USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/
```

```
CREATE OR REPLACE EDITIONABLE TRIGGER "TRG_MASK_PATIENT_PASSWORD"
BEFORE INSERT OR UPDATE ON PATIENT
FOR EACH ROW
BEGIN
  -- Ensure the password is masked before storing
  IF :NEW.PATIENT_PASSWORD IS NOT NULL THEN
    :NEW.PATIENT_PASSWORD := RPAD('*', LENGTH(:NEW.PATIENT_PASSWORD) - 3, '*') ||
      SUBSTR(:NEW.PATIENT_PASSWORD, -3);
  END IF;
END;

/
ALTER TRIGGER "TRG_MASK_PATIENT_PASSWORD" ENABLE
/
```

Column Name	Data Type	Nullable	Default	Primary Key
PATIENT_ID	NUMBER(12,0)	No	-	1
PATIENT_NAME	VARCHAR2(100)	Yes	-	-
PATIENT_IC	VARCHAR2(15)	Yes	-	-
PATIENT_PNO	VARCHAR2(20)	Yes	-	-
PATIENT_ADDRESS	VARCHAR2(200)	Yes	-	-
PATIENT_GENDER	VARCHAR2(10)	Yes	-	-
PATIENT_PASSWORD	VARCHAR2(20)	Yes	-	-

Figure 8.4.6 Table Patient

8.4.7. Table Appointment

```
CREATE TABLE "APPOINTMENT"
(
  "APPOINTMENT_ID" NUMBER(4,0),
  "APPOINTMENT_DATE" DATE,
  "STAFF_ID" NUMBER(4,0),
  "PATIENT_ID" NUMBER(12,0),
  "APPOINTMENT_TIME" VARCHAR2(15) COLLATE "USING_NLS_COMP",
  CONSTRAINT "APPOINTMENT_ID_NN" CHECK (appointment_id IS NOT NULL) ENABLE,
  CONSTRAINT "APPOINTMENT_DATE_NN" CHECK (appointment_date IS NOT NULL) ENABLE,
  CONSTRAINT "APPOINTMENT_ID_PK" PRIMARY KEY ("APPOINTMENT_ID")
  USING INDEX ENABLE
) DEFAULT COLLATION "USING_NLS_COMP"
/
ALTER TABLE "APPOINTMENT" ADD CONSTRAINT "APPOINTMENT_PATIENT_FK" FOREIGN KEY ("PATIENT_ID")
REFERENCES "PATIENT" ("PATIENT_ID") ENABLE
/
ALTER TABLE "APPOINTMENT" ADD CONSTRAINT "APPOINTMENT_STAFF_FK" FOREIGN KEY ("STAFF_ID")
REFERENCES "STAFF" ("STAFF_ID") ENABLE
/

CREATE OR REPLACE EDITIONABLE TRIGGER "APPOINTMENT_ID_TRIGGER"
BEFORE INSERT ON appointment
FOR EACH ROW
BEGIN
  -- Assign the next value of the sequence to the staff_id column
  :NEW.appointment_id := appointment_id_seq.NEXTVAL;
END;

/
ALTER TRIGGER "APPOINTMENT_ID_TRIGGER" ENABLE
/
```

Column Name	Data Type	Nullable	Default	Primary Key
APPOINTMENT_ID	NUMBER(4,0)	No	-	1
APPOINTMENT_DATE	DATE	Yes	-	-
STAFF_ID	NUMBER(4,0)	Yes	-	-
PATIENT_ID	NUMBER(12,0)	Yes	-	-

Figure 8.4.7 Table Appointment

8.5 DATA MANIPULATION LANGUAGE (DML)

8.5.1 TABLE PATIENT

```
INSERT INTO PATIENT (PATIENT_ID, PATIENT_NAME, PATIENT_IC,
PATIENT_PNO,      PATIENT_ADDRESS,      PATIENT_GENDER,
PATIENT_PASSWORD)
VALUES (001, 'Hwang Yeji', 000524040282, 0182218815, '10 JLN PPR
PAYA RUMPUT, MELAKA', 'FEMALE', 'YejiHwang00');
```

```
Insert into patient (PATIENT_ID, PATIENT_NAME, PATIENT_IC, PATIENT_PNO, PATIENT_ADDRESS, PATIENT_GENDER, PATIENT_PASSWORD)
values (001, 'Hwang Yeji', 000524040282, 0182218815, '10 JLN PPR PAYA RUMPUT, MELAKA', 'FEMALE', 'YejiHwang00');
```

EDIT	PATIENT_ID	PATIENT_NAME	PATIENT_IC	PATIENT_PNO	PATIENT_ADDRESS	PATIENT_GENDER	PATIENT_PASSWORD
	8787	Milia	021789028787	13213213	KUALA LUMPUR	F	MILIA
	2	Aqil	214112124	431431	LOT 455 TAMAN CEMPAKA	F	Aqil11
	1234	Aidil	123412	42135133	TAMAN INDAH MELAKA	M	AIDIL@
	1	Hwang Yeji	000524040282	182218815	10 JLN PPR PAYA RUMPUT, MELAKA	FEMALE	YejiHwang00

UPDATE PATIENT

SET PATIENT_PNO = 0192217714

WHERE PATIENT_NAME = 'Hwang Yeji';

```
update PATIENT
set PATIENT_PNO = 0192217714
where patient_name = 'Hwang';
```

EDIT	PATIENT_ID	PATIENT_NAME	PATIENT_IC	PATIENT_PNO	PATIENT_ADDRESS	PATIENT_GENDER	PATIENT_PASSWORD
	8787	Milia	021789028787	13213213	KUALA LUMPUR	F	MILIA
	2	Aqil	214112124	431431	LOT 455 TAMAN CEMPAKA	F	Aqil11
	1234	Aidil	123412	42135133	TAMAN INDAH MELAKA	M	AIDIL@
	1	Hwang Yeji	000524040282	192217714	10 JLN PPR PAYA RUMPUT, MELAKA	FEMALE	YejiHwang00

8.5.2 TABLE STAFF

```
INSERT INTO STAFF (STAFF_ID, STAFF_NAME, STAFF_PNO,
STAFF_ADDRESS, STAFF_USERNAME, STAFF_PASSWORD,
STAFF_ROLE, STAFF_DOB, STAFF_HIREDATE, DEPARTMENT_ID,
SUPERVISOR_ID, STAFF_AGE)
VALUES (null, 'Haziq Aqmal', 018379247, '12 JALAN SERIBU BUNGA,
MELAKA', 'aHaziq93', 'haziq1993A', 'NURSE', '03/12/1993', '01/30/2022',
5, null, null)
```

```
INSERT INTO STAFF (STAFF_ID, STAFF_NAME, STAFF_PNO, STAFF_ADDRESS, STAFF_USERNAME, STAFF_PASSWORD,
STAFF_ROLE, STAFF_DOB, STAFF_HIREDATE, DEPARTMENT_ID, SUPERVISOR_ID, STAFF_AGE)
VALUES (NULL, 'Haziq Aqmal', 018379247, '12 JALAN SERIBU BUNGA, MELAKA',
'aHaziq93', 'haziq1993A', 'NURSE', '03/12/1993', '01/30/2022', 5, null, null)
```

				MELAKA								
	134	Haziq Aqmal	18379247	12 JALAN SERIBU BUNGA, MELAKA	aHaziq93	haziq1993A	NURSE	03/12/1993	01/30/2022	5	-	-

```
SELECT *
FROM STAFF
WHERE DEPARTMENT_ID = '2';
```

```
select *
from staff
where department_id = '2'
```

Results	Explain	Describe	Saved SQL	History								
STAFF_ID	STAFF_NAME	STAFF_PNO	STAFF_ADDRESS	STAFF_USERNAME	STAFF_PASSWORD	STAFF_ROLE	STAFF_DOB	STAFF_HIREDATE	DEPARTMENT_ID	SUPERVISOR_ID	STAFF_AGE	
133	SYURAH	0129475295	NO. 23 JALAN BATU, KUALA LUMPUR	Syurah94	Syurah@94	NURSE	04/04/1994	12/02/2012	2	-	-	
130	MUHAMMAD	01283746288	51, TAMAN BARU, SELANGOR	MUHAMMAD	Muhammad@12	DOCTOR	12/01/1994	05/02/2010	2	-	-	
128	Aqil	0174292024	12, JALAN PERMAI, MELAKA	AqilAsyraf	Aasyraf@	DOCTOR	01/01/2025	01/23/2025	2	-	-	
3 rows returned in 0.01 seconds Download												

8.5.3 TABLE ADMIN

```
INSERT INTO ADMIN (STAFF_ID, ADMIN_START_DATE)
VALUES (131, SYSDATE)
```

```
INSERT INTO ADMIN (STAFF_ID, ADMIN_START_DATE)
VALUES (131, SYSDATE)
```

Table		
Data		
Indexes		
Model		
Constraints		
Grants		
Statistics		
UI Defaults		
Triggers		
Dependencies		
SQL		
REST		
Sample Queries		
Query		
Count Rows		
Insert Row		
Load Data		
EDIT		
STAFF_ID		
ADMIN_START_DATE		
	131	01/17/2025

```
INSERT INTO ADMIN (STAFF_ID, ADMIN_START_DATE)
VALUES (132, '02/11/2022')
```

```
INSERT INTO ADMIN (STAFF_ID, ADMIN_START_DATE)
VALUES (132, '02/11/2022');
```

Table		
Data		
Indexes		
Model		
Constraints		
Grants		
Statistics		
UI Defaults		
Triggers		
Dependencies		
SQL		
REST		
Sample Queries		
Query		
Count Rows		
Insert Row		
Load Data		
EDIT		
STAFF_ID		
ADMIN_START_DATE		
	131	01/17/2025
	132	02/11/2022
Download		

```
DELETE FROM ADMIN
WHERE STAFF_ID = 132
```

```
DELETE FROM ADMIN
WHERE STAFF_ID = 132
```

Table		
Data		
Indexes		
Model		
Constraints		
Grants		
Statistics		
UI Defaults		
Triggers		
Dependencies		
SQL		
REST		
Sample Queries		
Query		
Count Rows		
Insert Row		
Load Data		
EDIT		
STAFF_ID		
ADMIN_START_DATE		
	131	01/17/2025

8.5.4 TABLE DOCTOR

```
INSERT INTO DOCTOR (STAFF_ID, SPECIALIZATION,  
ROOM_NUMBER, MED_LICENSE_NUM)  
VALUES ('130', 'PHARMACIST', 'A1', '2010130')
```

```
INSERT INTO DOCTOR (STAFF_ID, SPECIALIZATION, ROOM_NUMBER, MED_LICENSE_NUM)  
VALUES ('130', 'Pharmacist', 'A1', '2010130')
```

	STAFF_ID	SPECIALIZATION	ROOM_NUMBER	MED_LICENSE_NUM
	130	Pharmacist	A1	2010130

```
UPDATE DOCTOR  
SET SPECIALIZATION = 'Dermatologist'  
WHERE STAFF_ID = '112'
```

```
UPDATE DOCTOR  
SET SPECIALIZATION = 'Dermatologist'  
WHERE STAFF_ID = '112'
```

EDIT	STAFF_ID	SPECIALIZATION	ROOM_NUMBER	MED_LICENSE_NUM
	142	Orthodontist	A2	2023142
	130	Pharmacist	A1	2010130
	112	Dermatologist	A2	2025112
	128	Pharmacist	A1	2023128

8.5.5 TABLE NURSE

```
INSERT INTO NURSE (STAFF_ID, CERTIFICATE_LEVEL)
VALUES ('136', 'Medical nursing')
```

```
INSERT INTO NURSE (STAFF_ID, CERTIFICATE_LEVEL)
VALUES ('136', 'Medical nursing')
```

EDIT	STAFF_ID	CERTIFICATE_LEVEL
	136	Medical nursing

```
UPDATE NURSE
SET CERTIFICATE_LEVEL = 'Licensed practical nurse'
WHERE STAFF_ID = '136'
```

```
UPDATE NURSE
SET CERTIFICATE_LEVEL = 'Licensed practical nurse'
WHERE STAFF_ID = '136'
```

EDIT	STAFF_ID	CERTIFICATE_LEVEL
	136	Licensed practical nurse
	150	Certified nursing assistant
	131	Advanced registered nurse
	134	Licensed practical nurse

8.5.6 TABLE DEPARTMENT

```
INSERT INTO DEPARTMENT (DEPARTMENT_ID,  
DEPARTMENT_NAME, DEPARTMENT_DESC)  
VALUES (NULL, 'Orthodontist', 'Orthodontist')
```

```
INSERT INTO DEPARTMENT (DEPARTMENT_ID, DEPARTMENT_NAME, DEPARTMENT_DESC)  
VALUES (NULL, 'Orthodontist', 'Orthodontist')
```

EDIT	DEPARTMENT_ID	DEPARTMENT_NAME	DEPARTMENT_DESC
	1	Dermatologist	Dermatologist
	61	Orthodontist	Orthodontist
	2	Pharmacist	Pharmacist
	3	Neurologist	Neuron
	4	Urologist	Urologist

```
UPDATE DEPARTMENT  
SET DEPARTMENT_ID = 5  
WHERE DEPARTMENT_NAME = 'Orthodontist'
```

```
UPDATE DEPARTMENT  
SET DEPARTMENT_ID = 5  
WHERE DEPARTMENT_NAME = 'Orthodontist'
```

EDIT	DEPARTMENT_ID	DEPARTMENT_NAME	DEPARTMENT_DESC
	1	Dermatologist	Dermatologist
	5	Orthodontist	Orthodontist
	2	Pharmacist	Pharmacist
	3	Neurologist	Neuron
	4	Urologist	Urologist

8.5.7 TABLE APPOINTMENT

```
INSERT INTO APPOINTMENT (APPOINTMENT_ID,
APPOINTMENT_DATE, STAFF_ID, PATIENT_ID)
VALUES (NULL, '01/12/2024', '1849')
```

```
INSERT INTO APPOINTMENT (APPOINTMENT_ID, APPOINTMENT_DATE, STAFF_ID, PATIENT_ID)
VALUES (NULL, '01/12/2024', '130', '1849')
```

EDIT	APPOINTMENT_ID	APPOINTMENT_DATE	STAFF_ID	PATIENT_ID
	19	01/12/2024	130	1849

Download

```
UPDATE APPOINTMENT
SET STAFF_ID = '142'
WHERE PATIENT_ID = '8787'
```

1 UPDATE APPOINTMENT

2 SET STAFF_ID = '142'

3 WHERE PATIENT_ID = '8787'

4

Results

Explain

Describe

Saved SQL

History

3 row(s) updated.

APPOINTMENT_ID	APPOINTMENT_DATE	STAFF_ID	PATIENT_ID
22	12/23/2024	142	8787
26	09/08/2024	130	1243
28	03/07/2025	130	1243
32	12/16/2024	112	1282
19	01/12/2024	130	1849
23	01/23/2025	142	8787
27	09/23/2024	130	1243
31	12/21/2024	112	9483
30	07/21/2024	112	9483
33	01/16/2025	112	1282
21	01/01/2025	130	1849

8.6 CD/FLASH DRIVE

Report submission via microsoft form provided by lecturer.

9.0 REFERENCES

Jabatan Kesihatan Negeri Melaka. (n.d.). *Jabatan Kesihatan Negeri Melaka*. Ministry of Health Malaysia. Retrieved December 23, 2024, from <https://jknmelaka.moh.gov.my>

10.0 RUBRIC

ICT502/ITS571 ITS571 PROJECT REPORT RUBRIC

ITEM	MARKS	GROUP
Table of Content (1 Mark)		CLASS:
INTRODUCTION		MEMBERS:
Company Background (2 Marks)		
2 Marks If the company background is presented		
CASE STUDY		
Problem Statement (5 Marks)		
1-3 Marks If they did not state that the current system is Manual or File-based Approach.		
4-5 Marks If they state that the current system is Manual or File-based Approach with some relevant sub problems because of the manual system.		
Objective (5 Marks)		
1-3 Marks If they state the system objective		
4-5 Marks If they state that they want to design, develop and test as the objective.		
SYSTEM DESIGN		
Flow Chart of System (10 Marks)		
1-5 Marks if there is flow chart but it is not reflecting the whole system		
6-10 Marks if the flowchart reflect the whole system		
10 SQL Queries (20 Marks)		
2 Marks for each query if they use different kind of SQL.		
1 Mark is for the repeated SQL		
For example:		
UPDATE EMP SET emplID = 100 WHERE name = 'Hamiz'; ← 2 Marks		
Update DEPT SET deptName = 'Finance' WHERE deptID = '10'; ← 1 Mark as the operation is almost the same as previous SQL.		

System Development Sample Screen (20 Marks)							
Read	Insert	Update	Delete	Bridge	Recursive	Inheritance	Extra
2 marks	2 marks	2 marks	2 marks	3 marks	3 marks	3 marks	3 marks
Extra can be anything related to database function. For example, use sequence for primary key (get 1 mark).						Total:	
Conclusion (5 Marks)							
5 Marks will be given if they have stated what is the conclusion from the project that have been developed.							
APPENDIX A: ERD							
20 Marks							
Rubric for ERD:							
Entity		4					
Attributes		3					
Relationship		3					
Relationship Name		2					
Cardinality/Modality		2					
Inheritance		2					
Recursive		2					
Bridge		2					
TOTAL		20 Marks					
APPENDIX B: Data Dictionary							
3 Marks will be given as long as it is being inserted in report							
APPENDIX C: DDL							
3 Marks will be given if all the DDL for ALL tables are presented.							
APPENDIX D: DML							
3 Marks will be given if all the DML for ALL tables are presented. At least 2 DML for each table.							
APPENDIX F: CD/FLASH DRIVE						TOTAL MARKS	
3 Marks will be given if they attached the CD/FLASH DRIVE at report during presentation.						100	

As a requirement, your group MUST use ORACLE for your database system.